Page No.
Date : / /

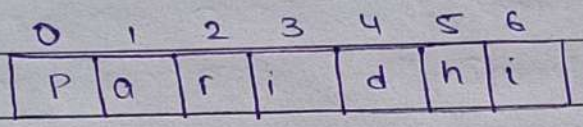
Lecture - 19

String

" paridhi"

String string_name;

cin >> string_name;



string_name[2] = 'r'

S = " Rohit"

M = "Mohit"

① T = S + M

output : RohitMohit

② T = S.append(M)

output = RohitMohit

S = "10";

T = '20';

S = S + T

ouput : 1020

string_name.size();

↓

output = 7

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S = " CoderArmy"

① S.pushback('s'); → Add s at last

S = CoderArmys

② S = S + 's';
S.pop_back();
↳ delete last letter

Sort:
sort(str.begin(), str.end())
str before sort : abchb
str after stort : abbch.

→ Code:

```
String str;
cin >> str;          // Input : paridhi
Cout >> str;         // output : paridhi
Cout << str.size();  // output : 7
Cout << str[1];      // output : a
```

→ String str1 = " paridhi singh"

```
Cout << str1;        // paridhi singh
Cout << str1.size(); // 13
```

→ String str2;

```
cin >> str2;         // paridhi singh
Cout << str2;        // paridhi
```

In input if we give space, tab then it can be terminate.

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If you want to consider space also then use getline :

```
String str;
getline (cin, str);      // Input : Paridhi singh
Cout << str;             output : paridhi singh
```

```
String str1, str2;
getline (cin, str1);     // Input : Paridhi singh
getline (cin, str2);     // Input : Riya Rani
Cout << str1 << " " <<str2;  output : paridhi singh Riya Rani
```

→ string str : "Ram is "a" boy";
↳ Given Error

→ string str : " Ram is \"a"\ boy";
↳ output : Ram is "a" boy.
\ → ognore next character and print as written.

→ string str : " Ram is ||a|| boy";
↳ output : Ram is |a| boy.

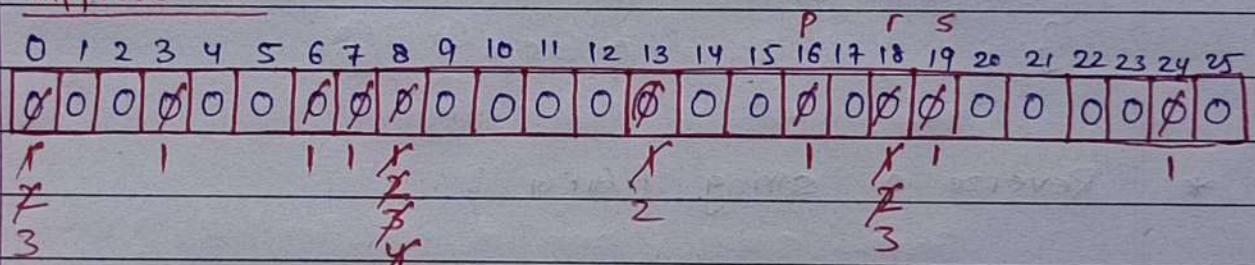
→ Char c = 'd' + 2;
Cout <<c; // output : f

* Sort by Comparing ASCII value:

```
for (int i=0;  i<n-1;  i++) {
    for (int j=1;  j<n;  j++) {
        if ( s[j] < s[j-1]) {
            swap (s[j], s[j-1]);
        }
    }
}
```

$O(N^2)$

* Approach - 2



s = pandhusinghRiyaRani

↳ Assum all are small

p a n i d h i s i n g h R i y a R a n i

a a a d g h i i i i i n n p r r r s y

↳ sorted.

Code:

```
String s;
cin >> s;
n = s.size();
int count[26];
for (int i=0;  i<26;  i++)
    count[i] = 0;
```

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```
for (int i=0; i<n; i++){
        int index = s[i] - 'a';
        count[index]++;
    }
    for (int i=0; i<26; i++){
        for (int j=0; j<count[i]; j++){
            char c = 'a' + i;
            cout<<c;
        }
    }
    return 0;
}
```

* Reverse a string (GFG):

```
p a r i d h i
↑               ↑
start           end
```

```
start < end
swap (p, i)
start++;
end--;
```

```
i a r i d h p
  ↑       ↑
start     end
```

```
start < end
swap (a, p)
start++;   end--;
```

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i h r i d a p

↑ ↑

start end

start < end

swap (r, d)

start++;

end--;

i h d i r a p

↑

start

end

start < end (false)

Code:

```
int start = 0;
int end = str.size()-1;

    while (start < end){
        swap (str[start], str[end]);
        start++;
        end--;
    }
return str;
}
```

Page No.:
Date: / /

* Anagram of string:

Example: S1 = cdd g k
S2 = gcd

output = 2

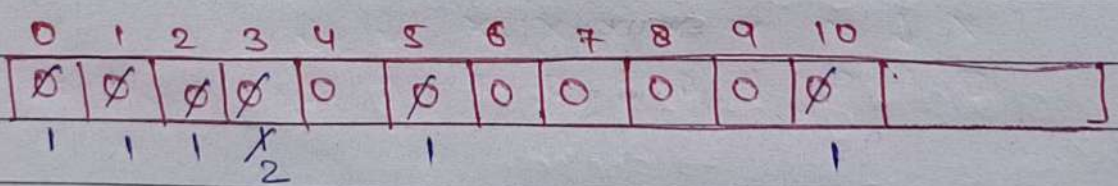
From S1 = one d is deleted
one k is deleted
total 2 operation.

Example:

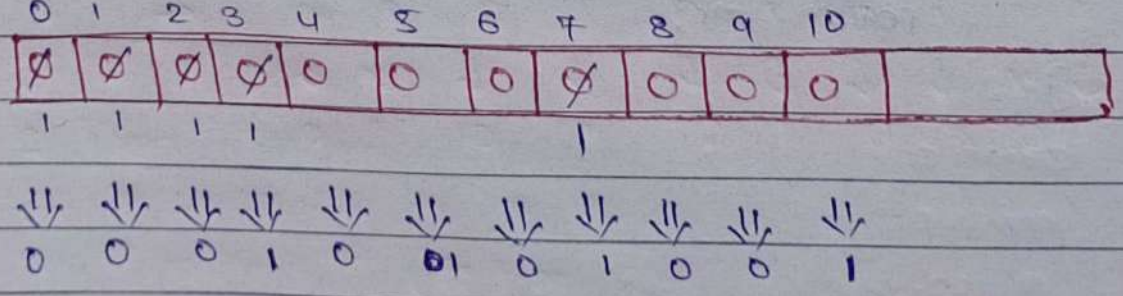
S1 = a b c d d k f
S2 = a b c d h

From S1: one d is deleted
one k is deleted
one f is deleted
S2: one h is deleted
4

S1



S2



SUM = 4

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Page No.:
Date: / /

* palindrome string:

Example = a b b a return 1
True

a b c return 0.
False

```
int  start = 0, end = s.size()-1;
     while (start <= end) {
          if (s[start] != s[end]) {
               return 0;
          start++;
          end--;
          }
     }
     return 1;
}
```

2nd Approach:

→ Reverse string and store in another string.
→ Match each character.

```
string k; k = s;
reverse (k.begin(), k.end());
     if (k == s)
          return 1;
     else
          return 0;
```

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* Minimum Number of Flips :

S = "001"

Alternate me alg alg character hona chaiye.

0 ke bagal me 1 होना
1 ke bagal me 0 hoga.

Hum ek-ek character ko idhar udhar kar sakte hai to humko kitna flip flop kar sakte hai.

Minimum flip return karna hai.

→ 0001
↑
1 0101 1 flip
return 1.

→ 0 0 0 1 0 1 0 1 1 1
↓ ↓ x x x x x x ↓ x
8 flips 1 0 1 0 1 0 1 0 1 0

→ 0 0 0 1 0 1 0 1 1 1
↓ x ↓ ↓ ↓ ↓ ↓ ↓ x ↓
2 flips 0 1 0 1 0 1 0 1 0 1

return 2

Page No.:
Date: / /

* Merge two string :

S1 : Hello

S2 : Bye

Ouptput : HBeylelo

```
string S3;
While (i < S1.size() && j < S2.size()){
        S3 = S3 + S1[i];
        S3 = S3 + S2[j];
          i++ ;
          j++;
}
While (i < S1.size()) {
        S3 = S3 + S1[i];
          i++;
    }
While (j < S2.size()) {
        S3 = S3 + S2[j];
          j++;
  }
```

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Page No.:
Date: / /

* To lower case:

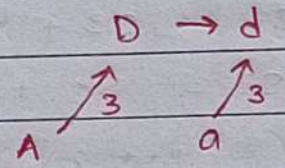
S1 = "LOVELY"

Output: lovely

S2 = "Here"

output: here

→ code:



char c = s[i] - 'A'

c = 'd' + s[i] - 'A';

if (s[i] >= 'A' && s[i] <= 'z')

* Sum of two large Number:

X = "25"

Y = "23"

Output = 48

X = "2500"

Y = "27"

Output = 2527

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int num = '7' - '0' = 7

int num = '0' - '0' = 0

carr = 0, num = 0

↳ sum Niklega.

Code:

```
int i = x.size() - 1;
int j = y.size() - 1;

String str = "";
int num, rem, carry = 0;
Char c;

While (i >= 0 && j >= 0) {
    int num1 = x[i] - '0';
    int num2 = y[j] - '0';
    num = num1 + num2 + carray;
    carry = num / 10;
    rem = num % 10;
    c = '0' + rem;
    str = str + c;
    j--;
    i--;
}
While (i >= 0) {
    int num1 = x[i] - '0';
    num = num1 + carry;
    carry = num / 10;
    rem = num % 10;
    c = '0' + rem;
    str = str + c;
```

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```
        i--;
    }
    While ( j>=0) {
        int num1 = y[j] - '0';
        num = num1 + carry;
        carry = num/10;
        rem = num % 10;
        c = '0' + rem;
        str = str + c;
        j--;
    }
    if(carry == 1)
        str = str + '1';

    int size = str.size()-1;
    while ( size > 0 && str[size] == '0'){
        str.pop_back(); size--;
    }
    reverse( str.begin(), str.end());
    return str;
}
}
```

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